

ADIT / GCET / MBIT
(CONSTITUENT INSTITUTIONS OF CVMU)
B. TECH – SEMESTER - 5 MID EXAMINATION – SEPTEMBER 2024
202045609 – MACHINE LEARNING

Date: 05/09/2024

Time: 11:30 PM TO 12:30 PM

Maximum Marks: 20

Instructions:

1. Figures on the right indicate full marks.
2. Make the suitable assumption if required, do specify the same.

Q. 1 (A) Answer the Following.

[04]

- (i) What is Reinforcement learning?
- (ii) Find the Interquartile Range (IQR) for the following dataset.

22, 25, 17, 19, 33, 64, 23, 17, 20, 18

- (iii) Describe what is Feature engineering in brief?
- (iv) List out various types of Regression techniques.

Q. 2 (A) Explain various categories of data in machine learning with suitable examples?

[04]

(B) How does K-fold cross validation method works? Explain in detail with its types.

[04]

OR

(B) Find the Accuracy, Precision, Recall and F1-score of a supervised learning model where the model predicted win and its actual won is 87, the model predicted win and its actual lost is 4, the model predicted loss and its actual won is 2, the model predicted loss and its actual lost is 7.

given x

[04]

Q. 3 (A) What is Ensemble model? How it can improve the performance of ML model? Explain the process in detail.

[04]

(B) What is Feature extraction? Write the steps of PCA algorithm used for feature extraction.

[04]

OR

(B) Find the R^2 error after applying the simple linear regression on the following dataset.

[04]

X	1	2	3	4	5
Y	3	4	2	4	5

***** ALL THE BEST *****

3.096